

Untitled1

May 18, 2026

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[4]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats

# 1. Load the dataset
df = pd.read_csv("children anemia.csv")

# Clean column names by stripping spaces and converting to lowercase
df.columns = df.columns.str.strip().str.lower()

# Map the exact column names printed from your directory
age_col = 'age in 5-year groups'
wealth_col = 'wealth index combined'
anemia_col = 'anemia level'

# Drop missing values across target columns
df_clean = df.dropna(subset=[age_col, wealth_col, anemia_col]).copy()

# =====
# ANALYSIS 1: Wealth Index Combined vs Anemia Level
# =====
print("===== ANALYSIS 1: WEALTH INDEX COMBINED VS ANEMIA LEVEL =====")
ct_wealth = pd.crosstab(df_clean[wealth_col], df_clean[anemia_col])
print("\n--- Contingency Table (Observed Frequencies) ---")
print(ct_wealth)

# Chi-square Test
chi2_w, p_w, dof_w, exp_w = stats.chi2_contingency(ct_wealth)
print(f"\nChi-square Statistic : {chi2_w:.4f}")
print(f"p-value : {p_w:.4e}")
print(f"Degrees of Freedom : {dof_w}")

# Contribution Calculation and Plot
exp_wealth_df = pd.DataFrame(exp_w, index=ct_wealth.index, columns=ct_wealth.
    ↪columns)
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contrib_wealth = ((ct_wealth - exp_wealth_df) ** 2) / exp_wealth_df

plt.figure(figsize=(8, 5))
sns.heatmap(contrib_wealth, annot=True, cmap="YlOrRd", fmt=".2f", cbar=True)
plt.title("Chi-square Contribution Diagram\n(Wealth Index Combined vs Anemia_
↳Level)", fontsize=11, fontweight='bold')
plt.ylabel("Wealth Index Combined")
plt.xlabel("Anemia Level")
plt.savefig('anemia_wealth_contribution.png', dpi=300, bbox_inches='tight')
plt.show()

print("\n" + "="*50 + "\n")

# =====
# ANALYSIS 2: Mother's Age vs Anemia Level
# =====
print("===== ANALYSIS 2: MOTHER'S AGE VS ANEMIA LEVEL =====")
ct_age = pd.crosstab(df_clean[age_col], df_clean[anemia_col])
print("\n--- Contingency Table (Observed Frequencies) ---")
print(ct_age)

# Chi-square Test
chi2_a, p_a, dof_a, exp_a = stats.chi2_contingency(ct_age)
print(f"\nChi-square Statistic : {chi2_a:.4f}")
print(f"p-value : {p_a:.4e}")
print(f"Degrees of Freedom : {dof_a}")

# Contribution Calculation and Plot
exp_age_df = pd.DataFrame(exp_a, index=ct_age.index, columns=ct_age.columns)
contrib_age = ((ct_age - exp_age_df) ** 2) / exp_age_df

plt.figure(figsize=(9, 6))
sns.heatmap(contrib_age, annot=True, cmap="Purples", fmt=".2f", cbar=True)
plt.title("Chi-square Contribution Diagram\n(Mother's Age vs Anemia Level)",
↳fontsize=11, fontweight='bold')
plt.ylabel("Mother's Age Groups")
plt.xlabel("Anemia Level")
plt.savefig('anemia_age_contribution.png', dpi=300, bbox_inches='tight')
plt.show()

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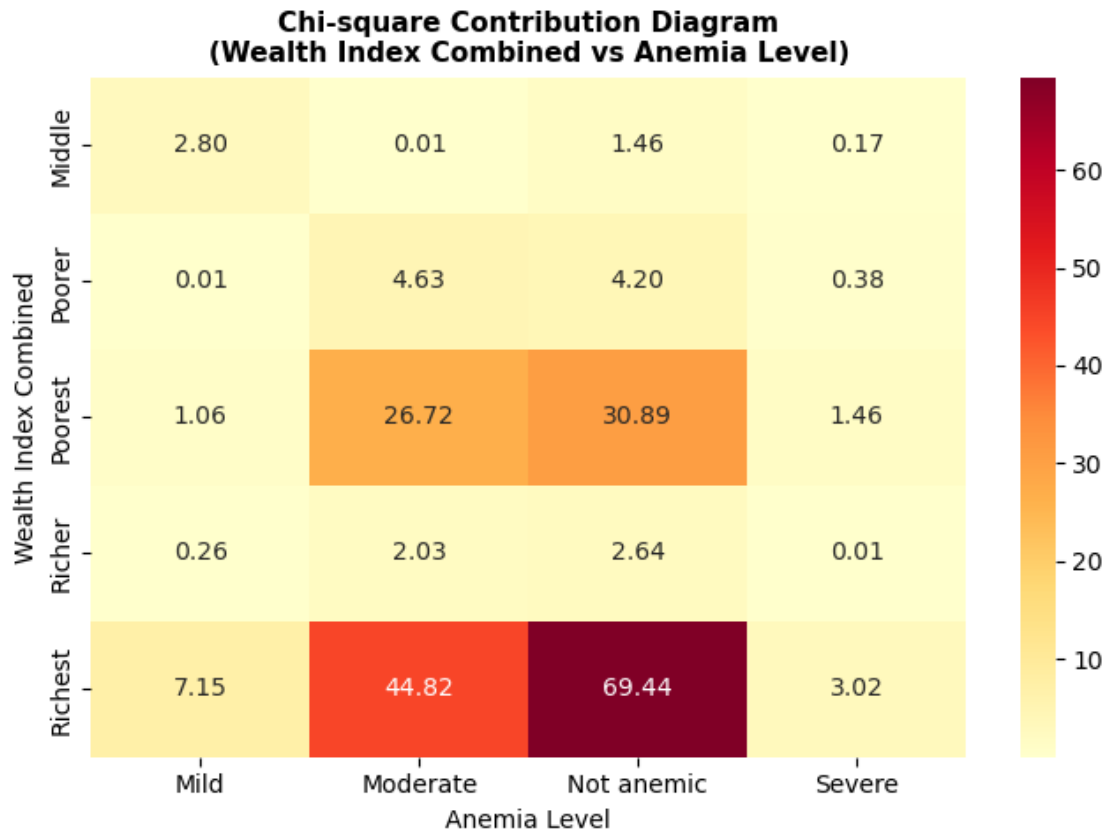
===== ANALYSIS 1: WEALTH INDEX COMBINED VS ANEMIA LEVEL =====

--- Contingency Table (Observed Frequencies) ---

anemia level	Mild	Moderate	Not anemic	Severe
wealth index combined				
Middle	840	874	1136	48
Poorer	745	883	1035	52

Poorest	784	985	936	57
Richer	721	772	1145	48
Richest	504	460	1085	26

Chi-square Statistic : 203.1542
p-value : 7.2798e-37
Degrees of Freedom : 12



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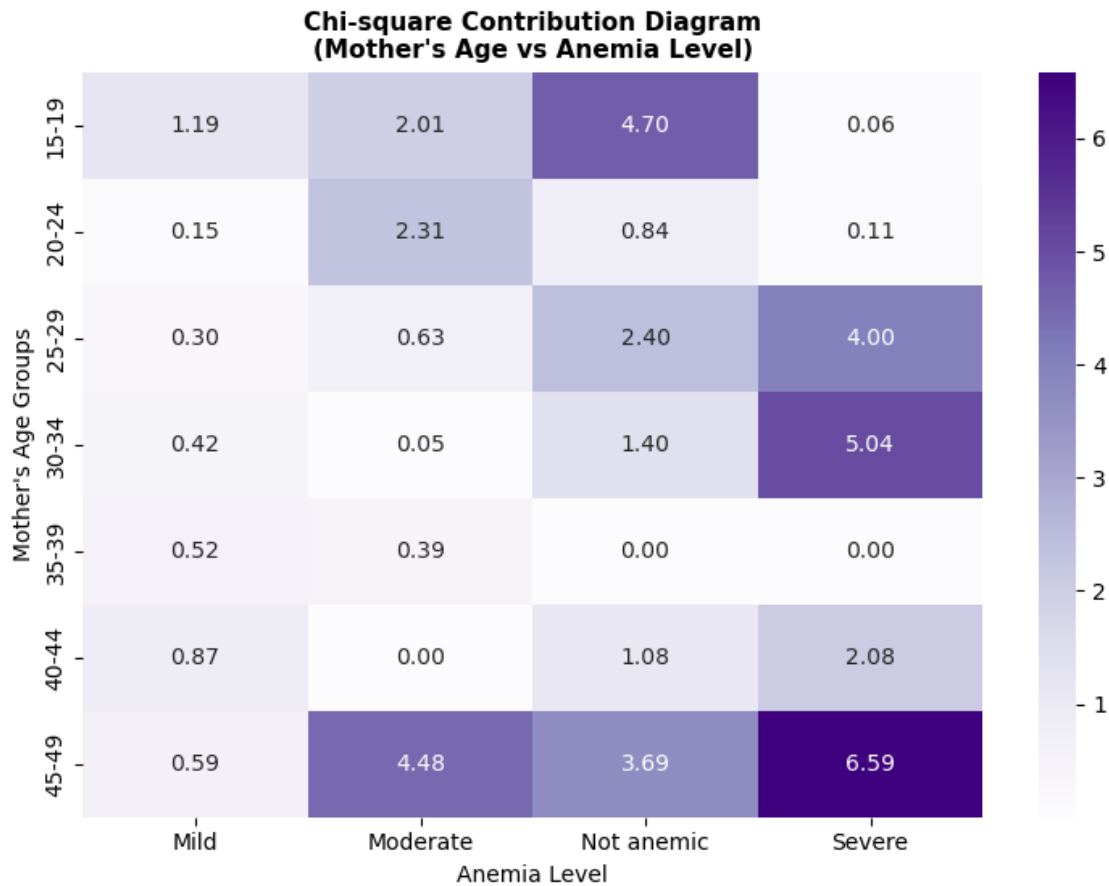
===== ANALYSIS 2: MOTHER'S AGE VS ANEMIA LEVEL =====

--- Contingency Table (Observed Frequencies) ---

anemia level	Mild	Moderate	Not anemic	Severe
age in 5-year groups				
15-19	157	177	182	10
20-24	630	748	922	39
25-29	996	1094	1565	49
30-34	852	928	1196	70
35-39	636	667	916	40

40-44	230	271	383	10
45-49	93	89	173	13

Chi-square Statistic : 45.8951
p-value : 3.0731e-04
Degrees of Freedom : 18



0.1 Problem 3: Chi-square Test of Independence (Children Anemia Level in Nigeria)

This analysis evaluates the structural dependencies regarding childhood anemia severity levels in Nigeria using the 2018 Nigeria Demographic and Health Surveys (NDHS) framework. The assessment explores the impact of household socioeconomic status and maternal age groups on the anemia profile of children aged 0–59 months.

0.1.1 Part 1: Socioeconomic Status (Wealth Index Combined) vs. Anemia Level

1. Hypothesis Formulation

- **H₀**: Childhood anemia levels are statistically independent of household socioeconomic status (Wealth Index Combined) in Nigeria.
- **H₁**: Childhood anemia levels are statistically dependent on household socioeconomic status (Wealth Index Combined) in Nigeria.

2. Chi-square Test Statistics

- **Chi-square Statistic (χ^2):** 203.1542
- **Degrees of Freedom (df):** 12
- **Asymptotic p -value:** 7.2798×10^{-37}

3. Statistical Interpretation The independence test evaluating socioeconomic hierarchies against childhood anemia severity generates a Pearson statistic of $\chi^2 = 203.1542$ with 12 degrees of freedom. The calculated asymptotic significance profile yields an extremely small value ($p = 7.2798 \times 10^{-37}$), which falls drastically below the standard $\alpha = 0.05$ critical threshold.

Consequently, the null hypothesis (**H₀**) is rejected with high empirical confidence. The analysis indicates a highly significant dependency between financial standing and childhood anemia vulnerability.

According to the mathematical cell contributions within the diagnostics diagram, the **Richest** cohort exhibits high positive variance towards remaining non-anemic (69.44 contribution), whereas the **Poorest** cohort shows structural vulnerability towards manifesting descriptive iron deficits (26.72 and 30.89 contributions).

0.1.2 Part 2: Maternal Age Groups vs. Anemia Level

1. Hypothesis Formulation

- **H₀**: Childhood anemia levels are statistically independent of maternal age groups in Nigeria.
- **H₁**: Childhood anemia levels are statistically dependent on maternal age groups in Nigeria.

2. Chi-square Test Statistics

- **Chi-square Statistic (χ^2):** 45.8951
- **Degrees of Freedom (df):** 18
- **Asymptotic p -value:** 3.0731×10^{-4}

3. Statistical Interpretation The statistical interaction assessment concerning maternal age cohorts and infant-to-child anemia categories establishes a Pearson test profile of $\chi^2 = 45.8951$ with 18 degrees of freedom. The resulting empirical significance value ($p = 0.000307$) is lower than the nominal testing alpha of 0.05.

Thus, we reject the null hypothesis (**H₀**), confirming that childhood anemia patterns vary across different maternal age categories.

The cell contribution mapping shows that the oldest reproductive category (**45–49** cohort) demonstrates distinct deviations, specifically within the “Severe Anemia” classification (6.59 contribution) and “Moderate Anemia” classification (4.48 contribution). Similarly, young maternal age (**15–19**

cohort) displays a distinct contribution signature regarding non-anemic outcomes (4.70). This indicates that physiological development, maturity, and specific caretaking tendencies across generational mothering groups act as compounding risk factors for childhood anemia.

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